

Debug Buddy

A guided debugging coach that helps learners find the bug themselves



Presented By:
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Ideathon | Developer Tooling | Debugging

The Problem

Beginners and students lose hours when their code does not work.



Errors written for experts

Error messages are written for experts. A learner reads them and still has no idea what to do.



Tools skip the learning

Most of the tools simply fix the code. The learner never builds debugging skill.



Shotgun debugging

Random edits and hope waste time and create new bugs.

Who is affected: students, bootcamp learners, junior developers

Our Idea

You paste buggy code. Debug Buddy does not reveal the bug. It guides you to find it.

- 1 **Plain-English summary:** what the code is trying to do
- 2 **Trace questions:** walk through the code by hand
- 3 **Gap analysis:** expected vs. actual, cause unnamed
- 4 **Zone and bug type:** e.g. loop setup, off-by-one; never the line
- 5 **Small experiments:** print a value, test with n=1
- 6 **Hint ladder:** 3 hints, vague to nearly there
- 7 **Last-resort reveal:** unlocked only after all hints



Core principle

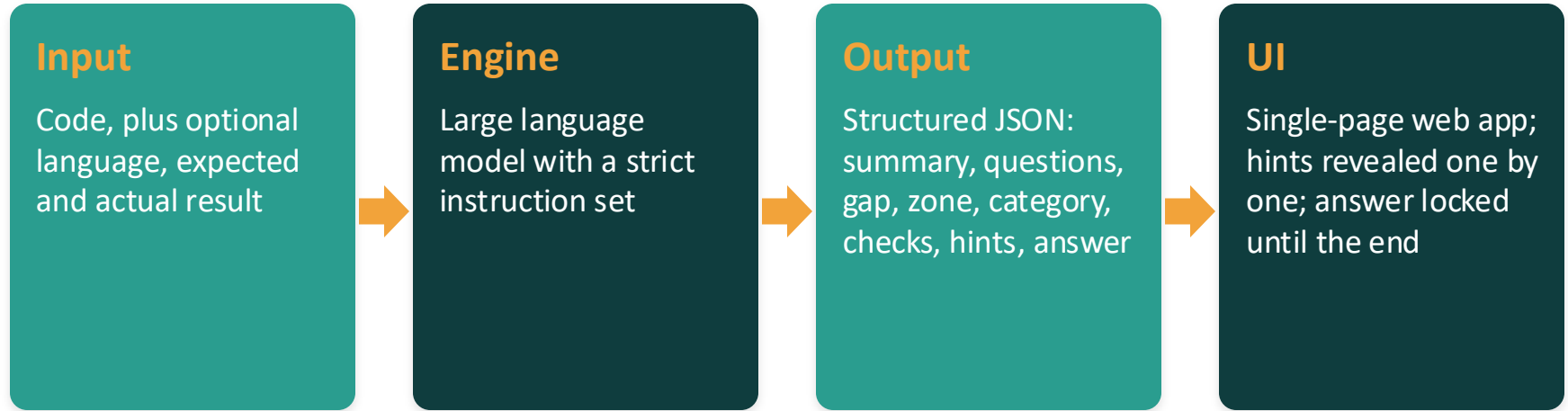
***Reduce wasted time
while increasing
understanding.***

How it is different

Other tools tell you the answer. Debug Buddy teaches you to find it.

	Typical tools	Debug Buddy
What you see	"Error on line 6"	"Look around the loop setup. What is the first and last value of i?"
The fix	Gives corrected code	Hints in steps; the learner makes the fix
Language	Technical jargon	Plain English
Approach	One-shot answer	Guided thinking process

Technical approach



Strict rules for the model

Never reveal the exact line

Never output fixed code

Use simple language

Feasible with what exists today



Works today

Off-the-shelf LLM plus a structured prompt; no custom model training.



Language-agnostic

Works for any programming language the model understands.



Low cost

One request per debugging session.



Prototype ready

A working single-page web app demonstrates the full hint ladder.

Impact



Less wasted time

Fewer hours lost to random trial-and-error.



Real skills

Builds debugging fundamentals, not dependence.



Less frustration

A calmer, more confident start for beginners.



Useful for teachers

Shows how students think and where they get stuck.

Roadmap across Developer Tooling

- 1 **Guided debugging with hint ladder**
- 2 Paste a diff: what changed, what could break?
- 3 Explain a file or folder in plain English
- 4 Run code, instant feedback on each hint-check
- 5 VS Code extension and API hooks
- 6 Simple language, dyslexia-friendly mode, multi-language

Debugging

Understanding Code Changes

Understanding Large Codebases

Developer Feedback Loops

API & Integration Tooling

Accessible Developer Tools

Phase 1 is the working prototype

Debug Buddy

Do not hand developers the fix. Teach them to find it.



[Live demo](#)